WebAP Dataset for IR Evaluation

BSU CS 437
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Version 1

This dataset has been adapted from the University of Massachusetts Amherst Web Answer Passages dataset found here.

The document collection consists of 6,377 web documents taken from the 2004 TREC Terabyte Track. Each document was parsed by UMA and tagged with sentence markers and relevance tags. In this adapted version the tags have all been stripped and the documents are plain text files (with no HTML or XML markup).

The original task for the WebAP collection was question answering and each document in the collection was judged to have a relevant sentence for a question. The questions were then summarized with a title. For example, the question for query 703 was:

```
What are the arguments the U.S. uses against joining the International Criminal Court?
```

The summary generated was:

```
U.S. against International Criminal Court
```

The summaries have been collected in the queries.txt file with their associated query id separated by the pipe symbol "|". For example, here are the first five lines from queries.txt:

```
701|U.S. oil industry history
702|Pearl farming
703|U.S. against International Criminal Court
704|Green party political views
705|Iraq foreign debt reduction
```

For each question the top 50 documents from a gold-standard search engine were collected. From these, human judges annotated relevant answers to the question. Note not all documents contained relevant answers.

The qrels.txt file lists the relevant documents for a given query id along with their relevance scale. For example, here are the first five lines from qrels.txt:

```
701 GX268-35-11839875 5
701 GX262-28-10569245 2
701 GX238-57-4348848 5
701 GX231-53-10990040 5
701 GX026-56-10685144 5
```

The first field is the query id, the second field is the document id, and the final field is the relevance scale from 1-5 where 1 is most relevant and 5 is least. The relevance scale corresponds to the judged answer sentence ranking from the original dataset.

The Python module webap.py included in this dataset has helper functions to work with the files.

`webap_rel_key(query_id, k=None, rel_file=f"{WEBAP_DIR}/qrels.txt")`

Returns a list of the top k (default all) relevant file ids for the given query_id.

`webap_query(query_id, query_file=f"{WEBAP_DIR}/queries.txt")`

Returns a tuple of the query id and query text for the given query_id parameter.

`webap_queries(query_file=f"{WEBAP_DIR}/queries.txt")`

Returns a list of all the queries in the form of (query id, query text) tuples